CASIE Dataset Preprocessing and Validation Report

Project: Cybersecurity Event Argument Extraction
Dataset: CASIE
Task: Dataset Inspection, Cleaning, Annotation Conversion, and Data Splitting

1. Objective

The objective of this work was to prepare the CASIE cybersecurity dataset for event argument extraction.

Unlike traditional named entity recognition, event argument extraction identifies the role of an entity or piece of information in a specific cybersecurity event. For example, a security report may describe an attacker, a victim, the tool used, and the data compromised.

The preprocessing pipeline was designed to preserve these relationships while ensuring that the dataset could be used reliably for model training and evaluation.

2. Initial Dataset Inspection

The original dataset contained 1,000 JSON documents with nested event and argument annotations.

The dataset was inspected for structural inconsistencies, invalid character offsets, duplicate documents, repeated annotations, and overlapping argument spans.

The structural validation did not identify missing required annotation structures. However, the offset validation identified 1,095 character-offset mismatches.

3. Annotation Cleaning

All 1,095 detected offset mismatches were corrected and subsequently validated.

The dataset was also examined for repeated and overlapping annotations.

One exact duplicate argument record was removed from the model-ready representation. Two annotations containing multiple valid argument roles were preserved because removing either role would result in information loss.

The original event-to-argument relationships were retained throughout preprocessing.

4. Duplicate Detection

Duplicate detection identified one pair of documents containing identical text and equivalent annotations.

Document `annotation/1554.json` was excluded, while `annotation/1530.json` was retained.

A second pair, `annotation/2663.json` and `annotation/891.json`, contained highly similar text but described different cybersecurity incidents.

Both documents were retained. However, a grouping constraint was introduced to ensure that they would be assigned to the same dataset split.

After deduplication, 999 documents remained.

5. Annotation Conversion and Tokenization

The nested CASIE annotations were converted into an event-based representation, producing one record for each event mention.

Each record preserved the source document, event identifier, event subtype, trigger, and associated arguments.

The documents were then tokenized and the character-level annotations aligned to word tokens.

Initial tokenization achieved 99.91% exact annotation alignment.

To resolve the remaining issues, annotation-aware tokenization was applied. This required splitting 23 tokens at annotation boundaries.

Two additional annotations contained trailing non-breaking spaces. These spaces were excluded from the effective token-alignment spans without changing the original annotation records.

Following these adjustments, all 31,176 trigger and argument annotations aligned successfully.

6. Event-Aware BIO Label Generation

Event-conditioned BIO labels were generated for all 8,482 event mentions.

The dataset contains 26 argument roles.

The labeling format distinguishes the beginning and continuation of argument spans using `B-ROLE` and `I-ROLE` labels.

Because multiple cybersecurity events can appear in the same document, labels were generated separately for each event rather than assigning one global set of labels to the document.

A multilabel representation was retained for annotations with simultaneous argument roles.

Final validation confirmed that all 22,694 argument records were represented and that all 8,482 event records passed the BIO integrity checks.

7. Training, Validation, and Test Splits

The dataset was divided at the document level using an approximately 80/10/10 distribution.

Split	Documents	Events	Arguments
Training	799	6,877	18,445
Validation	100	869	2,140
Test	100	736	2,109
Total	**999**	**8,482**	**22,694**

Document-level splitting ensured that events from the same source document were not distributed across different subsets.

The known near-duplicate pair was also assigned to the same split.

All five event subtypes and all 26 argument roles were represented in training, validation, and testing.

8. Validation and Quality Assurance

The final integrity checks confirmed that:

- All 999 documents were assigned to exactly one dataset split.
- All 8,482 event records were preserved.
- All 22,694 argument records were preserved.
- All exported event records matched the validated source records.
- No document overlap was detected across the three splits.
- The known near-duplicate grouping constraint was satisfied.
- The excluded exact duplicate was absent.
- All event-conditioned BIO records passed validation.

SHA-256 checksums were generated to support file integrity verification.

9. Known Limitations

Although the preprocessing pipeline passed its structural and alignment checks, several limitations remain.

First, some argument roles are rare. For example, the `Damage-Amount` role has only 42 examples across the entire dataset. Consequently, performance estimates for these categories may be unstable.

Second, annotation-aware tokenization introduced a small number of token fragments. Models using subword tokenizers will require appropriate alignment between the dataset's character offsets and the model's tokenization.

Third, event-conditioned labels require the model to distinguish arguments belonging to different events within the same document. A conventional document-level BIO sequence is insufficient to represent all event relationships.

Finally, the validation process verified structural consistency and annotation alignment, but it did not establish that every original annotation was semantically correct.

10. Final Deliverables

Two archives were prepared.

CASIE_Team_Dataset.zip contains the training, validation, and test files, dataset metadata, split information, checksums, and usage documentation.

CASIE_Preprocessing_Audit.zip contains the cleaning, validation, duplicate-detection, alignment, and distribution reports.

Both archives passed ZIP integrity verification.

11. Conclusion

The CASIE dataset was successfully inspected, cleaned, converted, tokenized, labeled, and divided into training, validation, and test subsets.

The final dataset contains 999 documents, 8,482 event mentions, and 22,694 argument records.

All detected offset inconsistencies were corrected, annotation-to-token alignment was completed, and event-to-argument relationships were preserved.

The resulting dataset is ready for the team's model development and evaluation stages, subject to the documented limitations.